

Phase 4A: Bucketing & Segmentation

Reflection Questions - Answers

1. Why do we convert continuous values into categories?

Raw numbers like `total_spend` are hard to act on directly. Turning them into labels like Gold, Silver, and Bronze makes the data usable for business decisions - a marketing team can say "target Gold customers with a loyalty offer," which is much simpler than reasoning about a raw dollar figure for every single customer.

2. What is the difference between business segmentation and technical bucketing?

Business segmentation starts from a policy someone has agreed on, for example "Gold means the customer has spent \$2000 or more." It stays fixed and is easy to explain. Technical bucketing (quantiles, `Bucketizer`, `percent_rank`) starts from the data itself and produces groups that are statistically convenient, such as equal-sized groups, but the boundaries shift automatically whenever the underlying data changes.

3. When would fixed thresholds fail?

Fixed thresholds fail when they are chosen without checking the actual data. For example, using a \$10,000 cutoff on a dataset where the highest customer spend is around \$8,300 results in zero Gold customers. Fixed thresholds can also become outdated over time - if average spend doubles next year, an old threshold may quietly stop meaning anything useful.

4. How does quantile-based segmentation differ from fixed rules?

Quantile-based segmentation always produces roughly equal-sized groups (for example, the top one-third, middle one-third, and bottom one-third of customers), regardless of the actual dollar values involved. Fixed rules keep the same dollar cutoff every time, which means group sizes can be very uneven, or even empty, depending on how the data is distributed.

5. Which method would you use in real-world projects?

Fixed business rules work best for customer-facing segments such as loyalty tiers, since customers and stakeholders need a stable, explainable definition. Quantile or percentile-based bucketing works best for internal analysis or machine learning features, such as identifying the top 10% of spenders in a given quarter, where consistent group sizes matter more than a specific dollar amount. In practice, many teams look at quantiles first to understand the data, then set fixed thresholds based on that analysis and keep them stable for a reporting period.